

A completeness hierarchy for multi-view interaction detection, and why naive multi-cohort application fails

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Abstract

Multi-view biomarker data, in which several partial measurements describe the same individual, are increasingly common in aging and disease research. A natural and widely entertained hypothesis is that interactions *between* views carry information about outcomes that the individual views do not. Organ-specific biological age gaps (BAGs) derived from plasma proteomics are a prominent example: it is tempting to ask whether the way two organs age *together* predicts disease beyond how each ages alone. We show that answering this question correctly requires more care than is usually applied, and we provide both the theory and a large-scale cautionary demonstration.

We define an operational hierarchy of multi-view combinations, from individual-view predictions and their disagreement, through cross-view products, to a joint nonparametric estimator, and prove three results. First, view disagreement, a popular multi-view summary, is provably blind to interactions: it is a function of main effects only, and its covariance with any interaction component is exactly zero under view independence. Second, a compression–order tradeoff bounds the highest interaction order recoverable from a given sample size and number of views. Third, when views are statistically dependent, for example through a shared batch factor, interaction signal leaks into individual-view predictions at a characterized rate, manufacturing apparent interactions where none exist.

We validate the hierarchy in simulation: with a planted interaction, cross-view products recover it while disagreement remains at chance, and the joint estimator degrades relative to structured products as dimensionality grows, exactly as predicted. We then apply the framework to the Global Neurodegeneration Proteomics Consortium (GNPC; nine validated organ BAGs, 18 disease outcomes). A naive pooled analysis appears to show that pairwise organ-aging interactions significantly improve disease prediction for all 18 outcomes. However, cross-cohort replication reveals Simpson’s paradox, the effect vanishes under cohort adjustment (mean incremental AUC $+0.020 \rightarrow -0.0006$), and a single-cohort analysis with in-cohort-trained BAGs yields null-to-negative increments. The apparent effect is entirely between-cohort batch structure, precisely the leakage regime our third theorem describes. We conclude with practical guidance: interaction detection in harmonized multi-cohort biomarker data is confounded by assay-batch structure unless explicitly controlled, and disagreement-based analyses cannot detect interactions at all.

1 Introduction

Biological age gaps quantify the difference between an organ’s predicted age, estimated from molecular or imaging features, and an individual’s chronological age. Organ-specific BAGs derived from plasma proteomics have emerged as informative markers of differential aging and disease risk [1–4]. As multi-organ panels become available, a compelling question follows naturally: beyond the contribution of each organ’s aging, does the *joint* pattern across organs, the interactions, carry additional information about disease? The framing is intuitive enough that it has been articulated directly in the field, that no organ system is an island [5].

This question is an instance of a general problem in multi-view learning, where each view is a partial, noisy observation of a shared system and the analyst wishes to know what the views reveal jointly that they do not reveal separately [6, 7]. Several heuristics are in common use. One especially intuitive approach is *view disagreement*: when the predictions from different views diverge for an individual, that divergence is taken as a signal, an idea with roots in co-training and multi-view consistency [8]. Another is to form explicit cross-view products and test whether they improve prediction.

Despite the appeal of these heuristics, the field lacks a rigorous account of what each can and cannot detect, and under what conditions they fail. This gap matters because the data on which these questions are increasingly asked, large harmonized consortia assembled from many contributing cohorts measured on different assay platforms, have exactly the structure most likely to produce confounded conclusions.

In this work we close that gap on both the theoretical and empirical fronts. We make four contributions:

1. **A hierarchy with completeness guarantees.** We define four operational levels of multi-view combination and characterize precisely what each captures (Section 2).
2. **A blindness result.** We prove that view disagreement is a function of main effects only and is uncorrelated with every interaction component, regardless of interaction order (Theorem 1). A widely used heuristic cannot, even in principle, detect the thing it is used to detect.
3. **A compression–order bound and a leakage characterization.** We bound the maximum interaction order recoverable from N samples and V views (Theorem 2), and we show that view dependence, such as a shared batch factor, causes interaction signal to leak into individual-view predictions at a characterized rate (Theorem 3).
4. **A large-scale cautionary demonstration.** Applying the framework to GNPC, we show that a striking apparent interaction effect across 18 diseases is entirely an artifact of between-cohort batch structure, reproduced from first principles in simulation and predicted by our leakage theorem.

The result is simultaneously a method, a set of guarantees about its components, and a warning: the most natural way to look for organ-interaction effects in multi-cohort proteomics will find them whether or not they exist.

2 The operational hierarchy

2.1 Setup

Consider V measurement views. View $v \in \{1, \dots, V\}$ observes features $X_v \in \mathbb{R}^{d_v}$, generated through a latent factor model

$$X_{v,j} = a_{vj}u_v + b_{vj}z + \sigma_v\varepsilon_{v,j}, \quad j = 1, \dots, d_v, \quad (1)$$

where $u_1, \dots, u_V$ are view-specific latent factors, z is a shared factor (e.g. chronological age), and $\varepsilon_{v,j}$ are independent noise terms. A scalar outcome depends on the view-specific latents through a function f :

$$Y = f(u_1, \dots, u_V) + \eta. \quad (2)$$

Under independence of the u_v , f admits the Hoeffding decomposition [9, 10]

$$f(u_1, \dots, u_V) = \mu + \sum_v f_v(u_v) + \sum_{v < w} f_{vw}(u_v, u_w) + \dots, \quad (3)$$

where each component f_S has the zero-marginal property $\mathbb{E}[f_S(u_S) \mid u_T] = 0$ for $T \subsetneq S$. The components with $|S| \geq 2$ are the *interactions*.

2.2 The four levels

We define four levels of combination, in increasing order of the structure they can capture:

- **L1 (linear)**. Best linear predictor in the original features. Captures linear projections of main effects.
- **L1' (view disagreement)**. Let $\hat{h}_v = \mathbb{E}[Y \mid X_v]$ be the view-specific optimal prediction. Disagreement is $D = \text{SD}(\hat{h}_1, \dots, \hat{h}_V)$, the spread across views in their individual predictions.
- **L2 (cross-view products)**. Individual views plus products of view summaries up to order p ; with mean summaries, the pairwise products are $\{\bar{X}_v \bar{X}_w\}_{v < w}$.
- **L3 (joint nonparametric)**. $\hat{Y}_{\text{np}} = \mathbb{E}[Y \mid X_1, \dots, X_V]$, estimated by k -nearest-neighbor or kernel regression on the concatenated features.

In the BAG application, each view is one organ, X_v is that organ's BAG, L1 is a model on the individual BAGs, L1' is the spread across organ BAGs, L2 adds pairwise BAG products, and L3 is a joint nearest-neighbor model. Figure 1 summarizes the hierarchy and what each level detects.

3 Theory

We state the three main results; full proofs are given in the Supplementary Material.

3.1 Disagreement is blind to interactions

Theorem 1 (Hierarchy completeness). *Under view independence: (a) L1 captures linear projections of main effects. (b) View disagreement L1' is a function of main effects only,*

$$D = \text{SD}(\mathbb{E}[f_v(u_v) \mid X_v])_{v=1}^V,$$

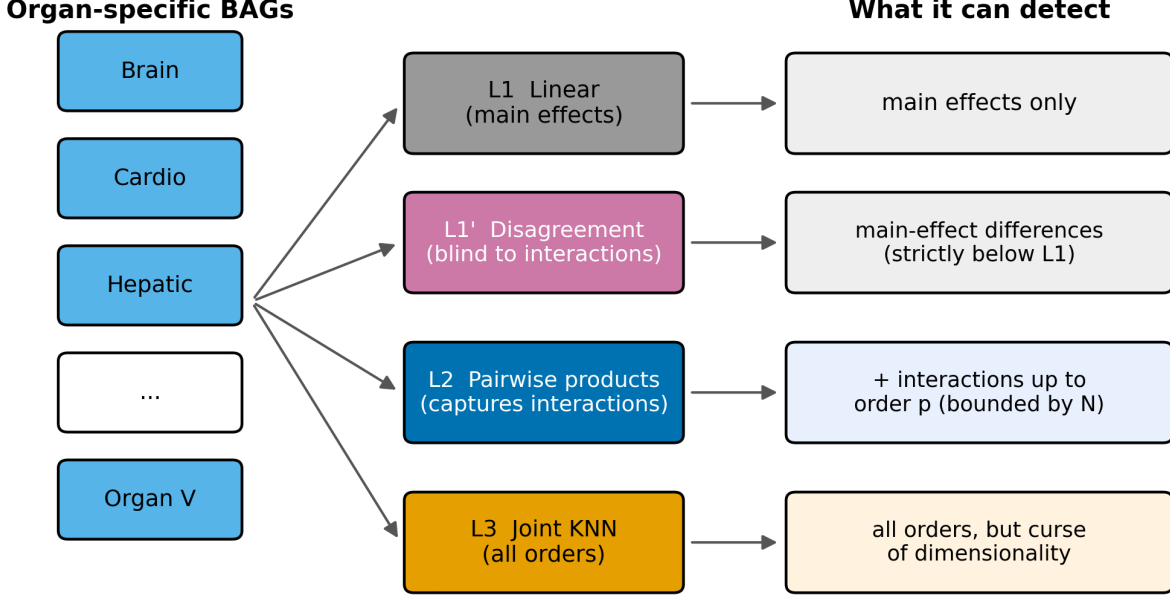


Figure 1: The operational hierarchy. Organ-specific BAGs feed four levels of multi-view combination. L1 (linear) captures main effects; L1' (disagreement) captures main-effect differences only and is provably blind to interactions; L2 (pairwise products) captures interactions up to an order bounded by sample size; L3 (joint nonparametric) captures all orders but suffers the curse of dimensionality.

and contains no interaction information: $\text{Cov}(D, f_S) = 0$ for all $|S| \geq 2$. (c) Order- p products (L2) capture interaction components up to order p , given interaction-sufficient summaries. (d) The joint estimator (L3) captures all interaction orders.

The key step is a lemma showing that the view-specific optimal prediction $\mathbb{E}[Y \mid X_v]$ depends only on the main effect f_v and is blind to every interaction f_S with $|S| \geq 2$, because the other latents are independent of X_v and the interaction terms have zero marginals. Since disagreement is a deterministic function of these view-specific predictions, it inherits their blindness. *A disagreement-based analysis therefore cannot detect cross-view interactions of any order.* This is not a matter of statistical power; the relevant covariance is exactly zero.

3.2 The compression–order tradeoff

Theorem 2 (Compression–order tradeoff). *With V views, homogeneous summaries of dimension q per view, and order- $\leq p$ products fit by penalized regression on N samples, the maximum recoverable interaction order satisfies*

$$p_{\max} \lesssim \frac{\log(N/\sigma^2)}{\log(Vq)}.$$

For polynomial interactions of degree m , the minimal sufficient summary has $q = m$, giving $p_{\max} \lesssim \log(N)/\log(Vm)$.

The number of order- p product features grows combinatorially, so the variance of the fitted model grows and the detectable order shrinks as V , q , or the required summary complexity increases.

A corollary is a *dimensionality crossover*: above a critical total dimensionality, the structured product estimator (L2) outperforms the joint nonparametric estimator (L3), which suffers the curse of dimensionality. This crossover is visible in both our simulations and the real data.

3.3 Leakage under view dependence

The independence assumption underlying Theorem 1 is an idealization. When views are correlated, the hierarchy boundaries become porous.

Theorem 3 (Leakage under dependence). *Suppose $\text{Cov}(u_v, u_w) = \rho$ for $v \neq w$, with standardized marginals, and Y contains main effects plus pairwise interactions. Then the conditional expectation of an interaction given one constituent view is nonzero,*

$$\mathbb{E}[u_v u_w \mid u_v] = \rho u_v^2,$$

so the view-specific prediction $\hat{h}_v$ acquires an interaction-contaminated term. The leakage correlation scales as $O(\rho^2)$ for Gaussian latents and $O(\rho^{3/2})$ for skewed latents. Consequently, products of dependent views can carry apparent interaction signal even when no interaction is present in f .

This is the theoretical heart of the cautionary tale. *When a shared factor couples the views, cross-view products pick up that shared factor rather than genuine interactions.* In multi-cohort data, the shared factor is assay batch: different cohorts measured on different proteomic platform versions have systematically shifted BAGs, and because disease prevalence varies across cohorts, the products correlate with disease through the cohort, not through biology. Section 5 shows this mechanism operating at scale.

4 Simulation validation

We first confirm that the hierarchy behaves as the theory predicts when its assumptions hold. We generated $N = 4,000$ samples with $V = 4$ independent latent organ factors, observed BAGs equal to the latents plus noise, and an outcome with main effects plus a single planted pairwise interaction (Methods).

The method recovers genuine interactions, and disagreement does not (Figure 2a). Cross-view products (L2, AUC 0.660) improved on individual views (L1, AUC 0.620) by 0.040, recovering the planted interaction. View disagreement (L1', AUC 0.518) remained essentially at chance, confirming Theorem 1's blindness result. The joint estimator (L3, AUC 0.623) captured the interaction but did not exceed the structured products.

Disagreement stays blind as the interaction grows (Figure 2b). Sweeping the interaction strength from zero to strong, the L2–L1 gain increased monotonically while the disagreement signal remained flat near chance. Disagreement does not merely underperform; it is structurally insensitive to interactions of any magnitude.

The compression–order crossover appears (Figure 2c). As the number of views increased, the joint nonparametric estimator (L3) fell progressively behind the structured product estimator (L2), consistent with Theorem 2's dimensionality crossover.

Together these establish that the hierarchy is a valid instrument: when interactions exist, L2 finds them, L1' cannot, and the structured estimator is preferred in higher dimensions.

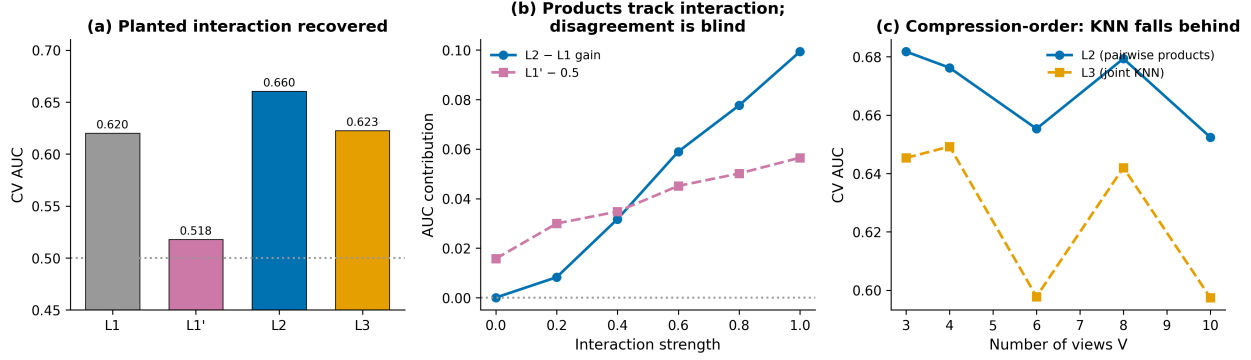


Figure 2: Simulation validation. (a) With a planted pairwise interaction, cross-view products (L2) recover it (+0.040 AUC over L1) while disagreement (L1') remains at chance. (b) As interaction strength increases, the L2–L1 gain rises while the disagreement signal stays flat, confirming structural blindness. (c) As the number of views grows, the joint estimator (L3) falls behind the structured product estimator (L2), the predicted dimensionality crossover.

5 Application to multi-cohort proteomics

5.1 Data and BAG derivation

We analyzed the GNPC v1 harmonized dataset, which integrates plasma proteomics and clinical phenotypes across 23 contributing cohorts measured on multiple SomaScan platform versions (v3, v4, v4.1). Following an established organ-aging framework [1, 3], we mapped tissue-enriched proteins to organ systems using the Human Protein Atlas [11] and trained organ-specific age-prediction models (LASSO, Elastic Net, linear SVR with nested cross-validation) on cognitively normal, EDTA-plasma, QC-passing participants, one sample per individual, with sex included as a covariate. Predicted-age bias was corrected using the Beheshti procedure fit on cognitively normal participants [12].

We retained the nine organ systems whose models genuinely learned age from proteins (Brain, Cardiovascular, Endocrine, Pulmonary, Immune, Hepatic, Pancreas, Digestive, Salivary). Five organ systems (Eye, Skin, Renal, and the sex-specific reproductive systems) were excluded: their bias-correction slopes were near -1 , indicating models that predicted approximately the cohort-mean age regardless of proteins, so their BAGs were noise rather than organ-specific aging signal. The disease analysis population in the current analysis comprised $N \approx 29,765$ records with both sexes represented. Because this count is consistent with the inclusion of longitudinal follow-up samples, the sample is being re-derived to a single baseline record per participant; the quantitative estimates reported below are therefore provisional pending that re-derivation. Eighteen disease outcomes with at least 50 prevalent cases were analyzed.

5.2 The naive pooled analysis appears strongly positive

For each outcome we compared L1 (the nine organ BAGs plus age and sex) against L2 (adding all 36 pairwise BAG products), using repeated stratified cross-validation and a paired test of the per-fold AUC differences, with Benjamini–Hochberg correction across outcomes. Pooling all cohorts, pairwise interactions significantly improved prediction for **all 18 outcomes** after FDR correction (Figure 3a). Incremental AUCs ranged from +0.034 (ALS) to +0.004 (CHF), with a mean of

approximately +0.020. The joint estimator (L3) was consistently lower than L2, reproducing the compression-order crossover in real data.

Taken at face value, this is a clean and exciting result: organ-aging interactions appear to carry disease information across a broad range of conditions. It would be straightforward, and wrong, to stop here.

5.3 Cross-cohort analysis reveals Simpson’s paradox

We recomputed the L2–L1 increment within each of the 11 cohorts containing at least 1,000 participants, and compared the within-cohort increments to the pooled estimate (Figure 3b). The pattern is diagnostic of confounding. For most outcomes the pooled increment exceeded nearly every within-cohort increment, and the within-cohort increments were frequently null or negative. For Alzheimer’s disease, every individual cohort showed a negative increment while the pooled estimate was positive, a textbook Simpson’s paradox. The ranking of which diseases showed the largest interaction gains did not replicate across cohorts: Spearman correlations between each cohort’s ranking and the pooled ranking ranged from 0.10 to 0.44 and none reached significance, and pairwise cohort agreements ranged from -0.70 to $+0.90$.

When the pooled effect is larger than its parts and does not replicate within parts, the effect lives in the between-group structure, not within individuals.

5.4 The effect vanishes under cohort adjustment

We repeated the hierarchy comparison adding cohort identity as a covariate (Figure 3c). The mean incremental AUC collapsed from approximately +0.020 to **-0.0006** , with several outcomes now showing significantly negative increments (L2 worse than L1). Notably, adding cohort sharply increased the L1 AUCs themselves (for example, Alzheimer’s from 0.72 to 0.85, Parkinson’s from 0.90 to 0.98), because diseases are unevenly sourced across cohorts and cohort identity is itself strongly predictive of diagnosis. With the between-cohort structure absorbed by the covariate, the products had nothing left to explain.

5.5 A single-cohort analysis confirms the null

The cleanest possible test removes batch structure entirely. We selected the cohort with both adequate cognitively normal participants (to train BAGs) and adequate disease cases (to test), retrained all organ BAGs using only that cohort, and tested the hierarchy within it (Figure 3d). With BAGs trained in-cohort and no cross-cohort structure of any kind, the mean within-cohort increment was **-0.022** . For Alzheimer’s disease, which had 1,291 cases and ample power, the increment was significantly negative (-0.008 , $p = 1.6 \times 10^{-4}$); Parkinson’s likewise (-0.024 , $p = 1.8 \times 10^{-5}$). Within a single homogeneous cohort, organ-aging interactions did not improve disease prediction; if anything, adding the product features degraded it, consistent with spending degrees of freedom on absent interactions, the other face of the compression-order tradeoff.

5.6 The mechanism, reproduced from first principles

To confirm that batch structure alone can manufacture the pooled effect, we simulated K cohorts, each with a different mean shift applied to all BAGs (an assay-batch effect) and a different disease

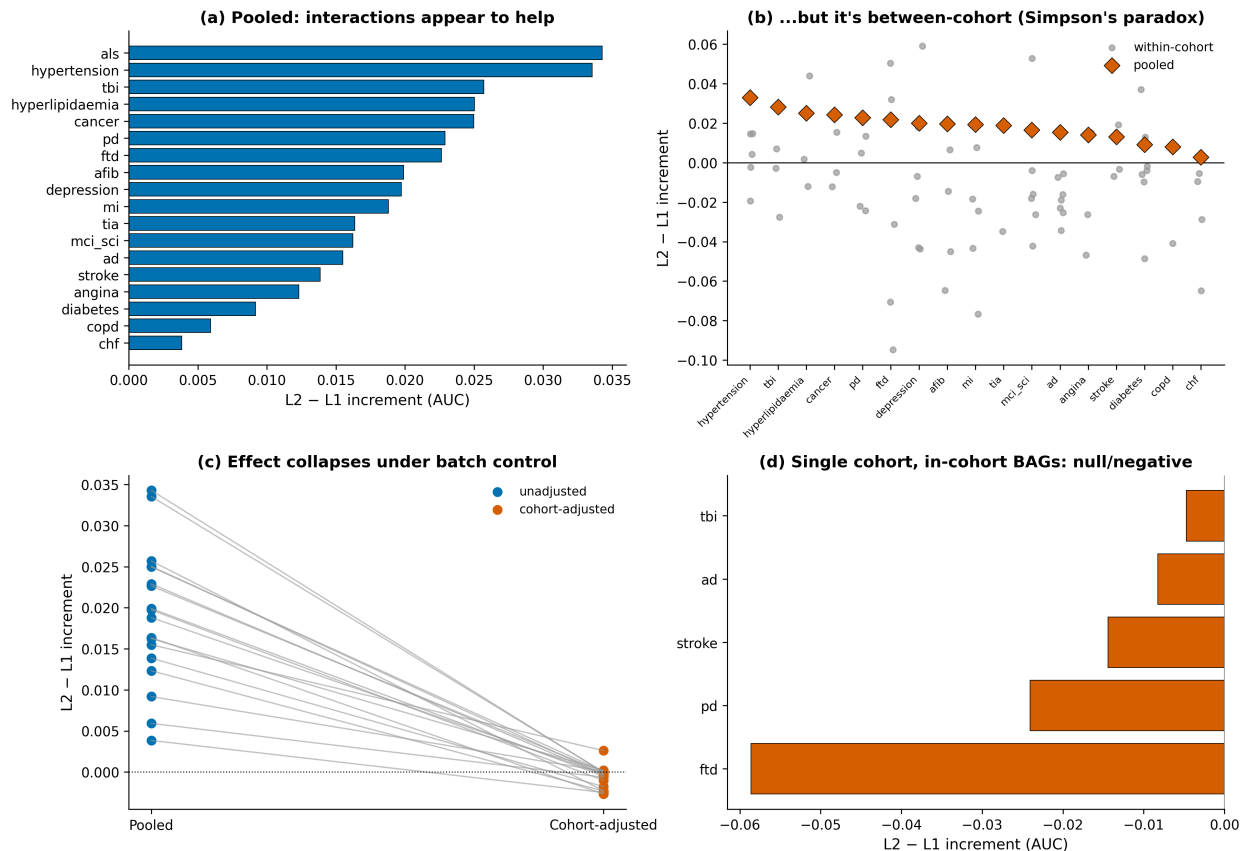


Figure 3: The GNPC interaction effect is a between-cohort artifact. (a) Pooled across cohorts, pairwise interactions appear to improve prediction for all 18 outcomes. (b) Within-cohort increments (grey) cluster near or below zero while the pooled estimate (orange) sits above them; for several outcomes every cohort is negative yet the pooled value is positive (Simpson’s paradox). (c) Adding cohort as a covariate collapses the mean increment from +0.020 to −0.0006. (d) In a single cohort with BAGs trained in-cohort, increments are null to negative, with Alzheimer’s and Parkinson’s significantly negative.

prevalence, with an outcome containing only main effects and no interaction whatsoever (Figure 4). As the batch-shift magnitude increased, the pooled L2–L1 gain rose above zero, becoming spuriously positive, while the mean within-cohort gain remained at or below zero. This is the GNPC pattern of Figure 3b regenerated from a model with no interactions, and it is exactly the leakage regime of Theorem 3: a shared factor coupling the views creates apparent interaction signal in the pooled data.

5.7 An apparent single-organ signal is also confounded

One outcome, Parkinson’s disease, showed an unusually high single-view AUC, with the Endocrine BAG alone separating cases at AUC 0.86 and Cohen’s d above 1.1. The endocrine protein set is dominated by pituitary and gonadal hormones (FSH, LH, CGA, PRL, GH1, POMC, TSHR), which vary strongly with age and menopausal status in ways a binary sex covariate does not fully capture. Removing the dominant organ reduced the full-model AUC only modestly (0.901 to 0.893), indicating the signal is distributed rather than driven by a single leaking protein, but the within-cohort and cohort-adjusted analyses show that even this distributed signal does not reflect organ-

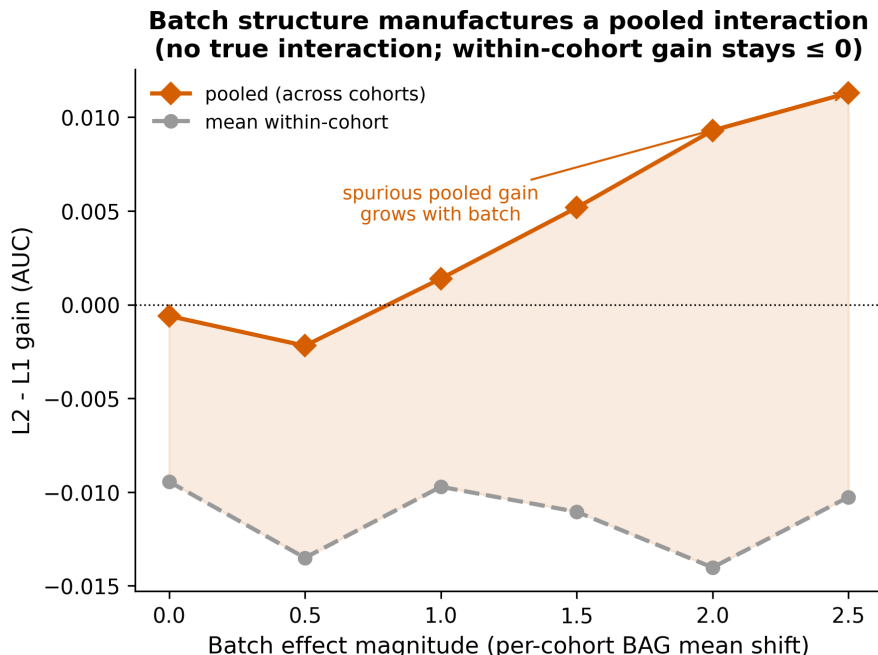


Figure 4: Batch structure manufactures a pooled interaction. Simulated cohorts with per-cohort BAG mean shifts and differing prevalence, with no true interaction. As the batch effect grows, the pooled L2–L1 gain becomes spuriously positive while the mean within-cohort gain remains at or below zero, reproducing the empirical pattern of Figure 3b and instantiating the leakage regime of Theorem 3.

interaction biology. We report it transparently as a demographic and batch-linked association rather than an organ-aging interaction.

6 Discussion

We set out to determine whether interactions between organ-specific aging clocks carry disease information beyond the organs’ individual contributions. The honest answer, in the largest harmonized proteomic dataset currently available, is no, and the more important contribution is the framework that explains why the question is so easy to get wrong.

A widely used heuristic cannot work. View disagreement is intuitive and common, but Theorem 1 shows it is a function of main effects only and is uncorrelated with every interaction. Studies that use disagreement to argue for interaction effects are measuring something other than interactions. If interactions are the target, explicit cross-view products (L2) or a joint estimator (L3) are required.

Multi-cohort interaction detection is confounded by default. Our central empirical finding is that a striking, broadly significant interaction effect across 18 diseases was entirely an artifact of between-cohort batch structure. The effect survived naive pooling and FDR correction; it did not survive cross-cohort replication, cohort adjustment, or a single-cohort retest. Theorem 3 explains the mechanism: harmonized consortia assemble cohorts measured on different assay versions, inducing shared batch factors that couple the views; because disease prevalence covaries with cohort, cross-view products proxy cohort and therefore disease. We reproduced this from first principles (Figure 4). Any multi-cohort study testing interaction effects without explicit batch control is at

risk of the same false positive.

Practical guidance. For interaction detection in multi-view, multi-cohort biomarker data we recommend: (i) never use disagreement as evidence of interaction; (ii) always include cohort or batch as a covariate, or analyze within batch and meta-analyze; (iii) treat a pooled effect that exceeds its within-cohort components as a confounding signature, not a discovery; and (iv) confirm any positive in at least one single-batch cohort.

Limitations. The GNPC BAGs were trained on pooled cognitively normal participants; while our single-cohort analysis retrains in-cohort, the primary analyses share this pooled-training step, which the cohort-adjusted and single-cohort results address but do not fully eliminate as a consideration. The diagnoses are prevalent rather than incident, so our analyses concern cross-sectional association rather than prediction of future disease. Our negative result pertains to this dataset and these organ definitions; we cannot exclude genuine organ-interaction effects detectable with incident outcomes, denser longitudinal sampling, or single-platform cohorts large enough to test interactions directly. Finally, our theory characterizes polynomial interactions through their conditional moments; non-polynomial interaction types (extreme-value, rectification) require summaries our framework does not yet specify, an open problem.

Conclusion. We provide a hierarchy for multi-view interaction detection with completeness guarantees, a proof that the popular disagreement heuristic is blind to interactions, a bound on the detectable interaction order, and a characterization of how view dependence manufactures false interactions. We validate the framework in simulation and demonstrate, at the scale of nearly thirty thousand participants, that a tempting organ-interaction effect is in fact a batch-confounding artifact predicted exactly by the theory. The method and the warning are inseparable: the most natural way to look for these effects will find them whether or not they are real, and only the structure provided here distinguishes the two.

Methods

Hierarchy estimation. L1 used penalized logistic regression on the individual view features plus covariates. L1' used the standard deviation across per-view predicted probabilities, each obtained by cross-validated logistic regression on a single view. L2 augmented L1 with all pairwise products of view summaries. L3 used k -nearest-neighbor classification on the concatenated standardized views ($k = 30$). Performance was the cross-validated area under the ROC curve. Increments were compared by repeated stratified k -fold cross-validation (5 folds, multiple repeats), with folds formed at the participant level so that no individual spans training and test, and a paired Wilcoxon test on the per-fold differences; multiplicity across outcomes was controlled by Benjamini–Hochberg FDR.

Simulation. Latent factors were standard normal; observed views were latents plus Gaussian noise (SD 0.3). Outcomes were Bernoulli with logit equal to main effects plus a planted pairwise product term whose coefficient set the interaction strength. The batch simulation drew, per cohort, a uniform mean shift applied to all views and a uniform intercept setting prevalence, with an outcome containing only main effects.

GNPC BAG derivation. Preprocessing retained EDTA-plasma, cognitively normal, QC-passing participants, restricted to a single baseline sample per participant, with no longitudinal follow-up samples entering model training or cross-validation; proteins were filtered to human analytes, log2-transformed, outlier-clipped, and grouped into organ systems by Human Protein Atlas tissue enrichment. For each organ, LASSO, Elastic Net, and linear SVR were trained with nested

cross-validation performed at the participant level and the lowest-test-MAE model retained; sex was included as a covariate. Beheshti age-bias correction was fit on cognitively normal predictions and applied to the disease population. Baseline-trained models may subsequently be applied to follow-up samples where a longitudinal question warrants it, separately from the cross-sectional analyses reported here. Organ models with bias-correction slope magnitude above 0.8 were excluded as non-informative. Disease labels were prevalent diagnoses coded as present versus absent.

Code and data availability. Analysis code is available at [repository URL]. GNPC data are available through the AD Workbench under consortium data-use terms.

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